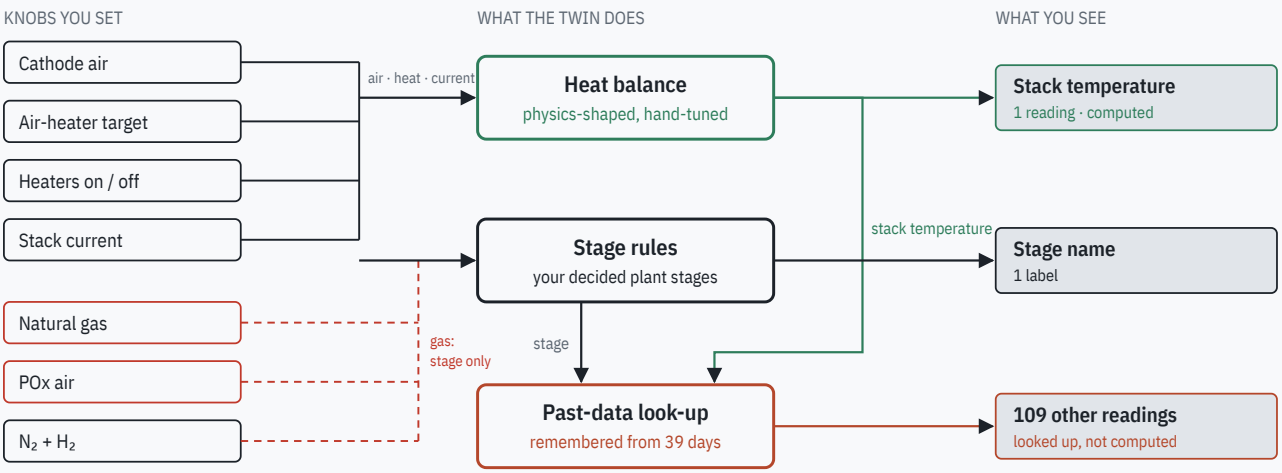
SOFC PLANT TWIN · PHYSICS CHECK

Does the twin follow the physics?

Partly. Only the stack temperature comes from a physics rule: heat in minus heat out. The other 109 readings repeat what the plant showed before in the same stage, so they do not react when you change a gas or air flow.

So the twin is good for **showing and explaining** the plant stages. It is not yet good for **“what happens if I change the gas flow?”** questions.



Natural gas, POx air and forming gas (red dashed) only decide the stage name. They never reach the heat balance, so no reading depends on how much gas flows.

Each part of the twin, in one line

Part of the twin	How it gets its number	Follows physics?
Stack temperature	One “heat tank”: heat from the air, the heaters and the current, minus heat lost through the walls. The numbers were tuned by hand to the measured warm-up and cool-down speeds.	Partly
Gas and air flows (6)	Read straight from your knobs.	Inputs
Stage name	Your decided plant-stage rules (air, gas, current, 210 °C).	Rules
109 other sensors	Looked up: “in this stage, at this stack temperature, the plant usually read …”. Fitted on 39 days.	No

Page 2 shows three tests the twin fails. Page 3 shows that the real plant itself does balance its energy, so a physics twin is possible. Page 4 is the plan to get there.

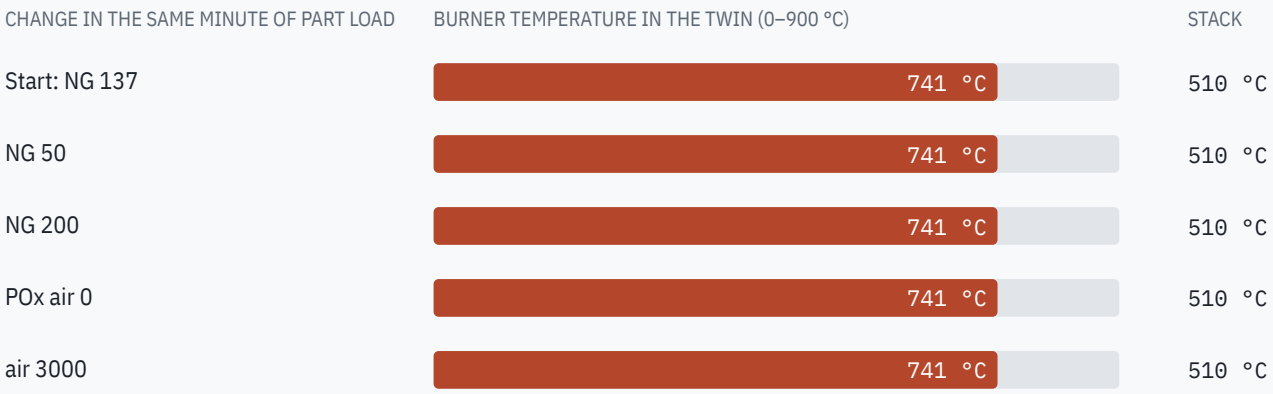
> IF YOU REMEMBER ONE THING
>
> Only one number in the twin is calculated; the other readings are remembered from the past.

Three tests the twin fails

A twin that follows physics must react when you turn a knob. These three tests use the twin itself and simple textbook numbers.

1 · Turn the gas knob: nothing moves

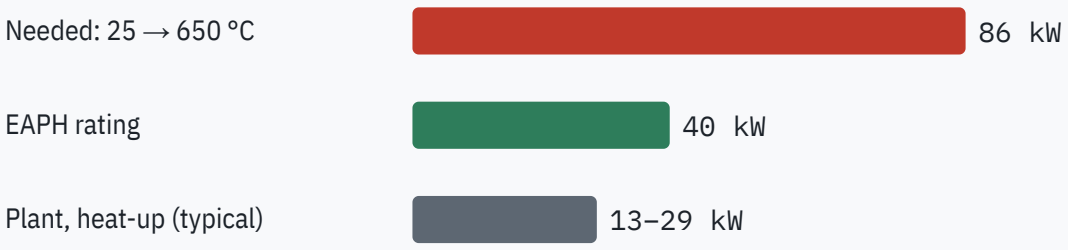
I put the twin in part load, then changed one knob at a time and read the sensors again.



Every case shows the same burner temperature (741 °C) and stack temperature (510 °C). Two hours later, with natural gas at 50 or at 137 ln/min, the stack is at 517 °C in both. In the real plant, less gas means less heat in the burner and a cooler outlet, and halving the cooling air makes the stack warmer.

2 · The air heater gets more power than it has

The cold-start button sets the electric air heater (EAPH) to 650 °C with 6000 ln/min of air. The twin simply lets the air reach that temperature.



Heat needed = air mass flow × heat capacity × temperature rise = 0.129 kg/s × 1.06 kJ/kg·K × 625 K ≈ 86 kW. The real plant manages because the air arrives pre-warmed by the two recuperators (APH, heat exchangers using the burner exhaust). The twin has no recuperators in its sums.

3 · More air cools faster than physics allows

Going from 6000 to 10000 ln/min is 1.67 times more air, so the air can carry at most about 1.67 times more heat. To match the measured cooling speeds, the twin needs 2.5 times more.



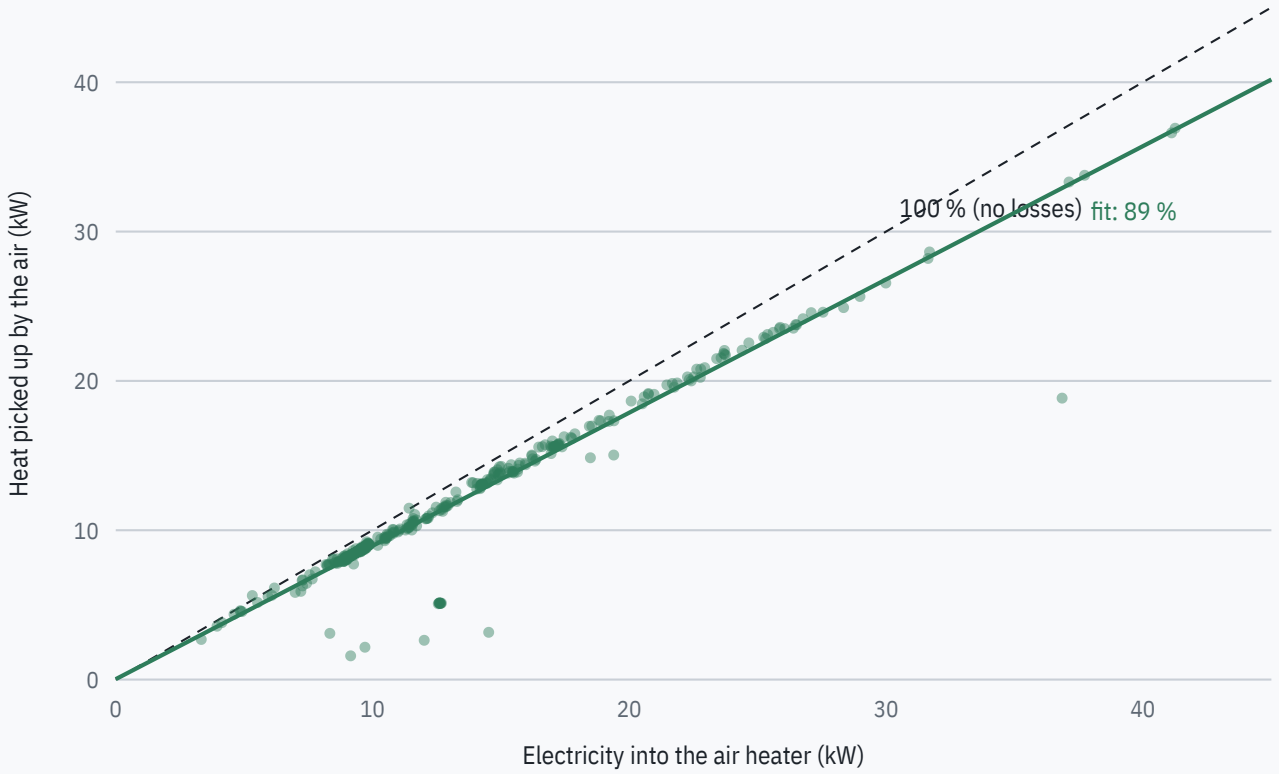
Something else changes when the plant blows strong air: for example the 3-way valve sending air past the recuperators, so cooler air reaches the stack. The twin hides that inside one tuned number.

IF YOU REMEMBER ONE THING

Change a gas flow in the twin and no reading changes. A physics twin would answer that question.

The real plant does follow the physics

Before building physics into the twin, I checked that the plant's own sensors obey a simple energy balance. They do, closely.



Each dot is 10 minutes of plant data (400 of the 2,798 windows drawn). Heat picked up by the air = air flow × heat capacity × (outlet − inlet temperature), from FIRC 1.1.1 + 1.1.2, TIR 2.05 and TIRC 2.06. Electricity = EAPH three-phase power.

What the numbers say

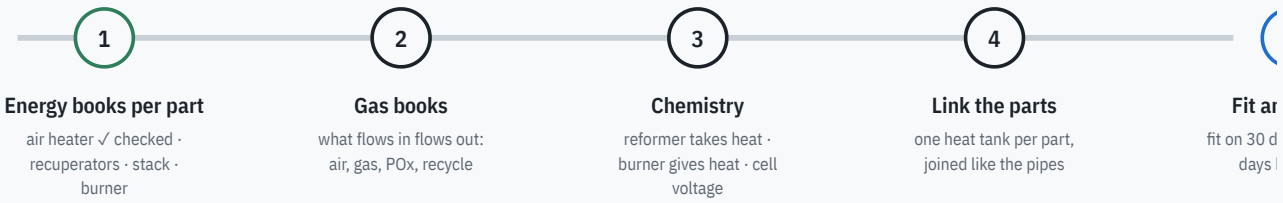
Measure	Value	Meaning
Share of electricity that ends up in the air	91 % (most windows 89–93 %)	About 9 % is lost through the heater casing, a normal amount.
How tightly the two follow each other	correlation 0.97	Almost a straight line: the sensors are consistent.
10-minute windows checked	2,798	Heat-up, hot standby and fuel switch.
Where it fits less well	part load 74 %, air-flow cooling 22 %	Only 18 and 15 windows, heater nearly off; treat as unreliable.

This matters because a physics twin is built the same way, one part at a time: **what goes in = what comes out + what is stored**. If the sensors could not balance even one heater, fitting physics would be hopeless. They can, so the plan on page 4 is realistic.

> IF YOU REMEMBER ONE THING
>
> The plant's air heater balances its energy within about 10 %, so physics can be fitted to this data.

Making the twin follow physics

Replace the “remembered” readings with small energy and gas balances, one part at a time, then tune them on the recorded days.



Same order as the earlier twin plan (heat tanks fitted to measured delays, then replay the recorded days). Step 1 has started: the air heater balance closes at 91 % (page 3).

What you would notice afterwards

TODAY

Gas knob changes the stage name only.
Air heater reaches any target.
Burner reads a stage average.

WITH PHYSICS

Less gas → less burner heat → cooler recuperator air.
Air heater limited to 40 kW; slower warm-up when air is cold.
Burner temperature follows the leftover fuel.

Decisions for you

Decision	Options	Suggestion
A · First scope	Heat-up, hot standby and cooling only (905 of 914 h of data) / also part load (4.5 h)	Heat stages first; part load has too little data to fit.
B · Accuracy goal	e.g. stack within ±20 °C on the held-back days	±20 °C stack, ±30 °C heaters and burner.
C · Sensors without physics yet	Hide them / keep the look-up, clearly labelled	Keep the look-up, labelled “typical, not computed”.
D · Order of parts	Air path first (air heater, recuperators, stack) / fuel path first	Air path first: it drives most of the warm-up and cool-down hours.

Back to the overview.

> IF YOU REMEMBER ONE THING
>
> Start with the energy books of the air heater, recuperators, stack and burner; that covers most of the plant's hours.